

Home Work 02: Linear Rational (Möbius) Quotient Derivative

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \frac{x-1}{x+1}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — LINEAR RATIONAL QUOTIENT FORMULATION

The problem consists of a first-degree rational polynomial function:

$$y = \frac{u(x)}{v(x)}, \quad \text{where } u(x) = x-1, \quad v(x) = x+1$$

Applying the standard Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}, \quad \text{with } \frac{d}{dx}(x \pm 1) = 1$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত অন্তরজ সূত্র (Standard Derivative)
লব (u)	$x-1$	$\frac{d}{dx}(x-1) = 1$
হর (v)	$x+1$	$\frac{d}{dx}(x+1) = 1$
হরের বর্গ (v^2)	$(x+1)^2$	$(x+1)^2$ (উৎপাদক আকারেই রাখা সমীচীন)

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

1. Sign Distribution in Subtraction: Pay close attention to sign distribution across the subtraction in the numerator:

$$(x+1) - (x-1) = x+1-x+1 = +2$$

Writing $x+1-x-1=0$ is a fatal sign distribution blunder.

2. Factored Denominator: Keep $(x+1)^2$ in factored form rather than expanding into x^2+2x+1 .

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{x-1}{x+1} \right)$$

ধাপ ২: ভাগফলের সূত্র (Quotient Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = \frac{(x+1) \frac{d}{dx}(x-1) - (x-1) \frac{d}{dx}(x+1)}{(x+1)^2}$$

ধাপ ৩: প্রমিত অন্তরজের মান $\frac{d}{dx}(x \pm 1) = 1$ বসিয়ে:

$$\frac{dy}{dx} = \frac{(x+1) \cdot 1 - (x-1) \cdot 1}{(x+1)^2}$$

ধাপ ৪: লবের বন্ধনী বিস্তার ও পদসমূহ কাটাকাটি (Cancellation) করে:

$$\frac{dy}{dx} = \frac{x + 1 - x + 1}{(x + 1)^2}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = \frac{2}{(x + 1)^2}$$

📖 ENGLISH NOTE — MONOTONICITY, RIGOR & SINGULARITY ANALYSIS

- **Domain of Function & Derivative:** Non-zero denominator requires $x + 1 \neq 0 \implies x \neq -1$. Thus, $\text{Dom}(y) = \text{Dom}\left(\frac{dy}{dx}\right) = \mathbb{R} \setminus \{-1\}$.
- **Strict Monotonicity:** For all $x \neq -1$, $(x + 1)^2 > 0$, which guarantees $\frac{dy}{dx} = \frac{2}{(x+1)^2} > 0$. The function is strictly increasing on both $(-\infty, -1)$ and $(-1, \infty)$ with no local extrema.

💡 ENGLISH NOTE — RAPID PARTIAL FRACTION / SHIFT VERIFICATION

Algebraic synthetic division prior to differentiation:

$$y = \frac{(x + 1) - 2}{x + 1} = 1 - \frac{2}{x + 1} = 1 - 2(x + 1)^{-1}$$
$$\frac{dy}{dx} = 0 - 2 \cdot (-1)(x + 1)^{-2} = \frac{2}{(x + 1)^2}$$

This verifies the Quotient Rule in under 5 seconds!

🎯 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Quotient Rule Formulation	Correct expansion of $\frac{vu' - uv'}{v^2}$	1.5
Polynomial Differentiation	Accurate derivative of $(x \pm 1) \implies 1$	1.5
Algebraic Simplification	Sign expansion $(x + 1) - (x - 1) = 2$	1.0
Final Standard Format	Writing $\frac{2}{(x+1)^2}$	1.0
Total AMIE Section-A Marks		5.0 / 5.0